Benjamin Goldstein

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EDUCATION

Georgia Institute of Technology College of Engineering — *Atlanta, GA*
Bachelor of Science in Electrical Engineering | GPA: 4.0 — *May 2029*

SKILLS

Hardware: STM32, Arduino, ESP32, Jetson Nano, Raspberry Pi, Stepper/DC/Servo Motors, Quadrature Encoders, Sensors, Soldering, Multimeter, Oscilloscope
Software: C, C++, Python, MATLAB, Java, RISC-V Assembly, ROS2, Linux, Git, GitHub, UART, Onshape CAD, KiCAD
Concepts: Embedded Systems, Feedback Control (PID), Digital Signal Processing, Heuristic Search, Machine Learning

PROJECT EXPERIENCE

STM32-Based Robotic Arm — *Agoura Hills, CA*
Independent Project — *May 2026 - Present*
- Designed a multi-DOF robotic arm integrating custom mechanical design, DC motors, and limit switch calibration
- Programmed embedded motor control firmware on STM32 implementing closed-loop DC motor PID position control
- Developed custom interrupt-driven UART protocol enabling reliable bidirectional communication between an STM32 microcontroller and NVIDIA Jetson Nano for ROS2-based motion control and teleoperation

Autonomous Connect 4 Robot — *Atlanta, GA*
Independent Project — *Sept 2025 - May 2026*
- Developed custom IR detection circuitry with digital EMA and first difference signal filtering for reliable move detection analyzed graphically in MATLAB
- Implemented PID-controlled piece placement and optimized bitboard game representation with finite-state control logic
- Designed custom UART protocol enabling reliable bidirectional communication between Raspberry Pi and Arduino
- Programmed pruned minimax-based AI engine to achieve optimal gameplay decision-making at a **7-move search depth** on Raspberry Pi Zero

Autonomous Rubik's Cube-Solving Robot — *Agoura Hills, CA*
Independent Project — *Aug 2024 - Aug 2025*
- Designed, fabricated, and integrated all mechanical, electronic, and software systems achieving **<1 min** autonomous solves
- Drove NEMA stepper motors with hall sensor feedback to ensure fast, accurate, and repeatable cube turning
- Built statistical color recognition algorithm using chi-square tests to reliably map cube faces under varied lighting conditions
- Implemented heuristic-driven depth-first search algorithm in Python on Raspberry Pi finding **≤35 move** solutions

Selected Hardware and Software Projects
Independent Projects
- Built a 9W spark-gap Tesla coil using flyback transformer circuitry, custom capacitor bank, generating **1-1.5 cm** plasma arcs
- Trained CNN skin lesion classifier on NVIDIA Jetson Nano, detecting malignant lesions with **~90% accuracy**
- Designed and prototyped tunable AM radio receiver using LC resonance, envelope detection, and analog audio amplification

EXPERIENCE

Autobots VIP (Vertically Integrated Project) — *Atlanta, GA*
VIP Team Member — *Jan 2026 - May 2026*
- Developed and debugged ROS2-based mobile robotics stack for Turtlebot2 and Kobuki platforms, integrating base drivers, LiDAR, and odometry across Gazebo simulation and hardware testing
- Implemented custom ROS2 control nodes in Python for Gazebo simulation, processing LiDAR and odometry data to enable reactive navigation behaviors (obstacle avoidance and PID motion control)

Medical Robotics Club — *Atlanta, GA*
Limbo Electrical Control Team Member — *Sept 2025 - Jan 2026*
- Contributed to PID control system design driving fingers, thumb, and wrist actuation in a multi-DOF robotic prosthetic hand

RELEVANT COURSEWORK

Intro Signal Processing, Hardware/Software System Programming, Digital System Design, Differential Equations, Honors Linear Algebra, Calculus I/II, Physics (Mechanics, E&M, Modern), Object Oriented Programming, Statistics